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Miniaturized Antenna Design for Wi-Fi 6E/Wi-Fi 7 Applications

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Abstract

To mitigate performance degradation caused by variations in host-platform ground configurations, this paper presents a compact wideband printed antenna for Wi-Fi 6E and Wi-Fi 7 applications. Occupying an ultra-miniature footprint of only 18 × 15 mm^2, the proposed antenna consists of an asymmetric T-shaped monopole (ATM), an open stub, and a meandered loop ground-line structure. The meandered ground line operates as a modified planar sleeve balun to reduce PCB-induced common-mode current propagation and improve robustness against representative host–ground variations. By controlling the electromagnetic coupling between the ATM radiator and the meandered ground loop, broadband impedance matching is achieved without additional matching circuits. The proposed antenna covers three operating bands: 2.4 to 2.5 GHz, 5.15 to 5.85 GHz, and 5.925 to 7.125 GHz. Measurement results demonstrate stable resonant frequencies and wide impedance bandwidths across the investigated host–ground configurations, with measured radiation efficiencies up to 88% at the investigated frequencies. The measured radiation patterns exhibit quasi-omnidirectional characteristics at 2.45 GHz, while multilobe behavior becomes more pronounced at 5.5 and 6.5 GHz due to higher-order modes associated with the meandered ground loop. A quantitative coverage-efficiency or minimum-realized-gain analysis is beyond the scope of this study. Although demonstrated using a USB wireless adapter, the proposed operating mechanism is associated with the antenna structure rather than the USB connector itself.

Keywords: Wi-Fi 6E; Wi-Fi 7 applications; ATM; sleeve-balun; USB wireless adapter

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1. Introduction

In recent years, to cater to aesthetic user preferences and the structural constraints of edge network devices, there has been an accelerating trend toward compact, lower-profile, and multi-band wireless communication products. Modern high-throughput frameworks, such as Wi-Fi 6E and the emerging Wi-Fi 7 standards, mandate that terminals concurrently support operations across three core frequency allocations: the 2.4 GHz band (2.4 to 2.5 GHz), the 5 GHz band (5.15 to 5.85 GHz), and the newly unlicensed 6 GHz band (5.925 to 7.125 GHz). However, the extreme spatial limitations encountered within modern terminal hardware have severely hampered the operating bandwidth, impedance matching stability, and total radiated efficiency of integrated radiating elements. In order to achieve miniaturization while maintaining multi-band resonance, numerous compact antenna architectures have been extensively investigated in the literature [1,2]. For instance, various

modern multi-antenna topologies, including two-port microstrip layouts with enhanced isolation [3], dual-band MIMO systems designed for smart glasses applications [4], and circularly polarized configurations utilizing defected ground structures (DGS) [5], have been presented to expand operating bandwidths under small-form-factor constraints. Furthermore, recent designs have explored tri-band inverted-F antennas with reduced ground support [6] and integrated metal-frame slot structures with tiny ground clearances [7] to accommodate the expanded Wi-Fi 6E/7 bands within narrow boundaries. Despite these geometric advancements, a critical and recurring bottleneck in miniaturized antenna deployment is their sensitivity to variations in the host–ground-plane configuration. Because a compact antenna cannot operate in complete isolation, it must interact with the metallic chassis or printed circuit board (PCB) of its accompanying device. When the electrical length of the antenna itself is inherently insufficient, the finite ground-plane of the device acts as an imperfect return path, generating strong common-mode currents along the host platform [3,7,8]. In certain design paradigms, utilizing a small antenna element allows the host–ground-plane to be beneficially exploited as an extended radiator to enhance overall gain and impedance bandwidth through ground radiation techniques, given that the chassis is typically much larger than the radiating element itself [9–12]. However, although incorporating the host–ground-plane into the total radiating system yields size advantages, it introduces severe operational vulnerabilities during practical deployment. Specifically, the antenna's input impedance, absolute gain, and 3D radiation patterns vary unpredictably depending on the specific dimensions and geometry of the host–ground-plane it interacts with [13]. Furthermore, in commercial industrial applications, achieving miniaturization often requires pairing specialized compact antennas with external lumped matching networks or tuning stubs [14,15]. Alternatively, advanced unified RF modules incorporating electromagnetic interference (EMI) suppression have been developed to shield against platform noise [16]. Nevertheless, because the antenna elements lack sufficient natural resonant lengths, their high-frequency performance remains heavily dependent on the ground-plane boundary conditions, often resulting in severe impedance mismatching or frequency offsets when migrated across different hardware platforms. To evaluate the proposed antenna under practical operating conditions involving different host-platform ground configurations, a Wi-Fi USB dongle was selected as the experimental validation platform. A USB dongle operates as a standalone wireless transceiver; however, during operation, it must be connected to a variety of host platforms, such as laptop computers and edge routers [10]. Consequently, the antenna is exposed to different ground-plane configurations and impedance environments, which may alter its input impedance and lead to resonant-frequency variations. It should be emphasized that commercial wireless products normally incorporate a dedicated RF front-end, including an RF transceiver IC, bandpass filters, and impedance-matching networks, between the digital interface and the antenna. In this work, however, the proposed antenna is intentionally implemented on a simplified USB dongle platform to represent a practical and challenging operating scenario. This simplified configuration enables the antenna performance to be evaluated under different host-platform ground configurations while minimizing the influence of additional RF front-end circuitry. To reduce the performance variations associated with different host-platform ground configurations, this paper presents an ultra-miniaturized printed tri-band antenna with an overall footprint of only 18×15 mm^2. The proposed topology integrates an asymmetric T-type monopole (ATM), a custom open stub, and a novel meandered loop ground-line structure extending from the main USB dongle body. By surrounding the active radiator, the meandered ground line functions as a modified planar sleeve balun that suppresses PCB-induced common-mode currents and reduces the electromagnetic interaction between the radiator and the host–ground plane [8,17,18].

A comprehensive analysis and validation of the underlying wave mechanisms, simplified boundary conditions, and rigorous multi-ground matrix evaluations are presented in this work, supported by a robust loop network and common-mode tracking framework [19,20]. Furthermore, while representative planar multi-band monopoles incorporating single-cell metamaterials or concentric ring-patch radiators loaded with defected ground structures (DGS) [21,22] have demonstrated effective multi-band excitation, they generally require larger radiator clearance areas and remain constrained by host–ground interactions. As a result, while the practical Wi-Fi 7 standard heavily relies on multi-antenna (MIMO) architectures, complex coexistence schemes, and high-throughput system-level integration, the proposed antenna exhibits improved robustness against variations in host-platform ground configurations. The proposed antenna is intended as a compact modular building block that can be readily extended to future multi-port antenna systems for Wi-Fi 6E and Wi-Fi 7 applications. The host-platform configurations investigated in this work are intentionally defined as controlled and representative finite-ground environments rather than as replicas of any specific commercial laptop, router, or other wireless product. This approach enables the influence of host–ground dimensions on the proposed antenna to be evaluated in a controlled and reproducible manner. It should be emphasized that the objective of this work is not to establish universal ground-plane independence or complete system-level robustness for all wireless terminals. Instead, the study focuses on the antenna-level design principle of integrating an asymmetric T-type monopole with a meandered sleeve-loop structure to reduce the sensitivity of a compact antenna to representative finite-ground variations. The host–ground configurations investigated herein are intentionally controlled to isolate the electromagnetic influence of ground-plane dimensions. Accordingly, the experimental results demonstrate the robustness of the proposed antenna under the investigated representative configurations, while broader validation involving realistic device geometries, complete system integration, polarization-resolved characterization, and multi-element MIMO operation remains outside the scope of the present study.

The main contributions and distinct novelties of this work are summarized as follows:

- Development of a Co-designed Radiator–Balun Architecture for Improved Platform Robustness: Unlike conventional compact multi-band antennas that rely on separate lumped-element matching networks or additional decoupling structures, the proposed topology integrates a boundary-grounded meandered loop directly with an asymmetric T-type monopole. The meandered loop performs dual functions as both a parasitic radiator for low-frequency impedance matching and a planar sleeve balun, thereby reducing the influence of host-platform ground variations without requiring additional decoupling networks.
- Establishment of a Multi-Ground Evaluation Methodology: Rather than optimizing the antenna for a single reference platform, this work introduces a multi-ground evaluation methodology that considers representative host-platform ground configurations during the design process. This approach enables a more comprehensive assessment of antenna robustness under practical operating conditions.
- Compact Tri-band Antenna for Wi-Fi 6E and Wi-Fi 7 Applications: The proposed antenna occupies only 18×15 mm^2 while covering the three principal Wi-Fi operating bands (2.4 GHz, 5 GHz, and 6 GHz). Broadband impedance matching, high radiation efficiency, and satisfactory omnidirectional radiation characteristics are achieved without additional matching circuits.
- Experimental Validation on a Practical USB Dongle Platform: The proposed antenna is implemented on a simplified USB dongle platform to evaluate its performance under representative host-platform ground configurations. Surface-current analysis and electromagnetic field distributions are further employed to illustrate how the pro-

posed structure reduces PCB-induced common-mode currents and improves robustness against platform variations.

The remainder of this paper is organized as follows. Section 2 presents the antenna design, current-distribution analysis, equivalent-circuit validation, and coupling-path investigation. Section 3 discusses the experimental results, performance evaluation, error analysis, and comparisons with related work. Finally, Section 4 concludes the paper. The USB wireless adapter is used only as an experimental validation platform because it provides a practical and challenging host–ground environment. The operating mechanism presented in this work is derived from the interaction between the asymmetric T-type monopole and the meandered ground loop, and therefore is independent of the specific USB interface implementation. The USB platform is adopted solely to experimentally evaluate the antenna performance under representative host-platform ground configurations.

2. Antenna Design and Current Path Analysis

2.1. Antenna Design and Structure

The geometrical configuration and detailed dimensions of the proposed tri-band antenna system are illustrated in Figure 1a. The total structure is implemented on a standard FR4 glass-epoxy substrate with overall board dimensions of 18 mm × 15 mm × 1.0 mm (W2 + W3 + W4 + GL1, L1, and thickness, respectively), featuring a relative permittivity of 4.3 and a loss tangent of 0.023. To satisfy the stringent reproducibility requirements of high-frequency radiating elements, the layout is designed as a single-sided planar structure where all radiating components, feedlines, and the meandered grounding system are fabricated on the top metal layer. As shown in Figure 1b, the total layout includes a small system ground plane (representing the USB dongle body) measuring 10 mm (GL1) × 15 mm (L1), and a protruding active radiating portion. Crucially, the active antenna radiator occupies an ultra-compact clearance area of only 8 mm (L2) × 15 mm (L1), making it highly suitable for space-constrained wireless communication environments. The detailed geometric parameters of the radiating branches and meandered lines are labeled in Figure 1b, and their corresponding physical dimensions are systematically cataloged in Table 1. The proposed antenna incorporates an asymmetric T-type monopole (ATM), an open stub, and a meandered loop ground-line structure. By establishing electromagnetic coupling between the ATM radiator and the surrounding meandered ground line (highlighted by the solid red circle in Figure 1b), the proposed structure supports three resonant modes covering the 2.4 GHz (2.4 to 2.5 GHz), 5 GHz (5.15 to 5.85 GHz), and 6 GHz (5.925 to 7.125 GHz) bands. In this architecture, the main monopole primarily determines the 6 GHz resonance, whereas electromagnetic coupling with the peripheral meandered loop facilitates operation in the 2.4 GHz and 5 GHz bands. Because the operating characteristics (e.g., resonant frequency, impedance bandwidth, and radiation pattern) of a compact radiator are influenced by the return-current path provided by the host platform, the antenna and its finite ground plane should be considered as an integrated system during the design process. The proposed structural integration enables the finite ground plane to cooperate with the meandered sleeve balun structure, thereby reducing PCB-induced common-mode currents and improving the antenna's robustness against variations in representative host-platform ground configurations.

Table 1. Geometrical parameters of the proposed miniaturized antenna.

Parameter Code	Description	Value (mm)
L1	Width of the meandered loop	15
L2	Length of the meandered loop	8
L3	Length of the asymmetric top-loading branch	4.5

Table 1. *Cont.*

Parameter Code	Description	Value (mm)
L4	Length of the open-stub branch	2
W1	Width of the asymmetric top-loading segment	2.5
W2	Width of the primary monopole feedline	1
W3	Trace width of the meandered-loop ground line	0.5
W4	Clearance width of the meandered-loop structure	6.5
W5	Spacing between meandered ground traces	1
GL1	Length of ground plane	10

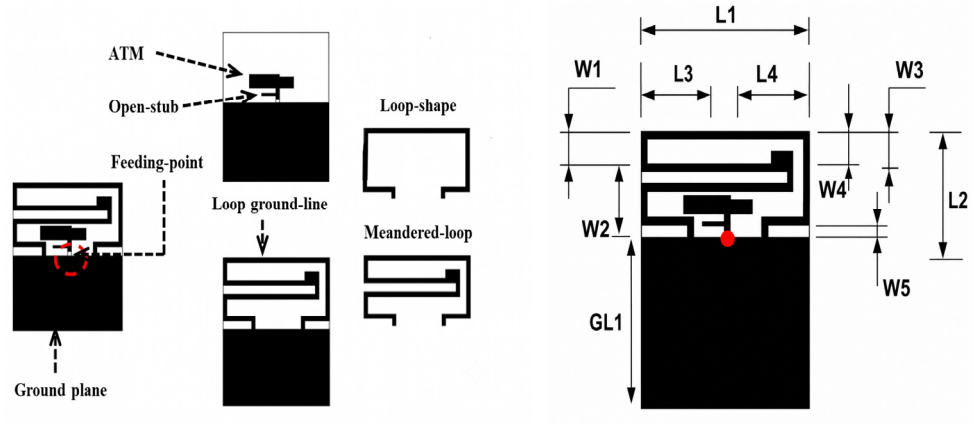


Figure 1. (**a**) Geometry of the proposed Wi-Fi triple-band antenna. (**b**) The detailed geometric parameters of the radiating branches and meandered lines, and the red circular indicator marks the explicit position of the feed point.

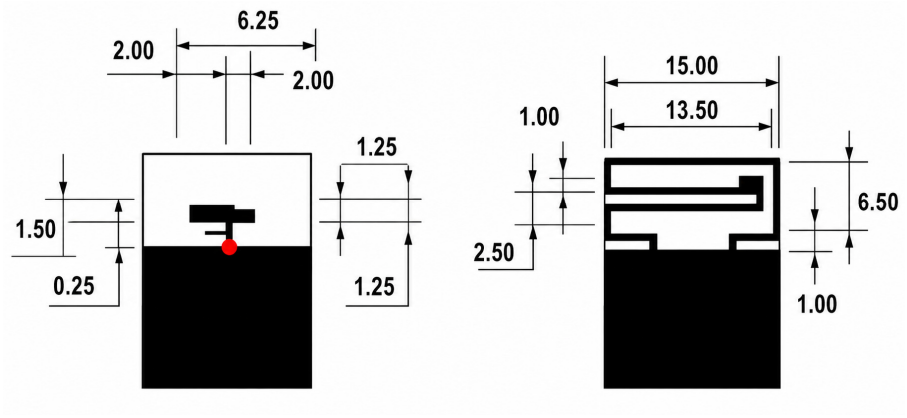
2.2. Antenna Design and Operation Mechanism

The proposed ATM structure consists of a monopole radiator and an asymmetric T-type top loading, as illustrated in Figure 2a. The monopole antenna is designed for operation in the 6 GHz band, and the electrical length of the radiating elements can be determined using one-quarter-wavelength at 6.5 GHz. The function of the asymmetric T-type top loading and open stub is to increase the impedance bandwidth. The meandered loop ground-line structure acts as a sleeve balun, with both the start and end points grounded. Based on the input impedance equation of a transmission line, shown in Equation (1), for multi-band operation requirements, the electrical length of the loop shape (outer loop structure) can be determined using one-quarter-wavelength at 5.5 GHz. The total electrical length of the meandered loop can be determined using one-quarter-wavelength at 2.45 GHz. This helps ensure that the input impedance of the meandered-loop ground-line structure is close to infinity, reducing the common-mode current induced by the PCB:

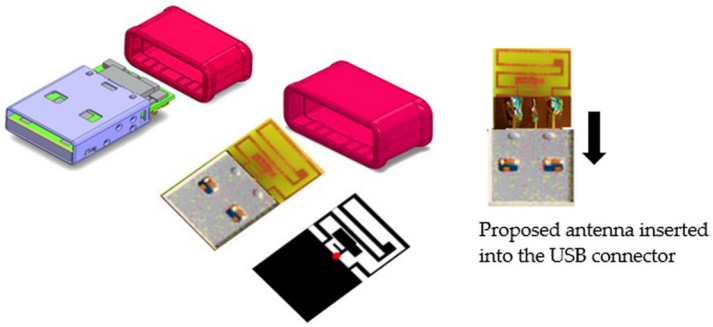
$$Z_{in}(l) = Z_o\left[\frac{Z_L + jZ_o\tan(\beta l)}{Z_o + jZ_L\tan(\beta l)}\right] \quad (1)$$

The spacing between the ATM and the ground-line structure, as shown in Figure 1a, is a critical parameter for controlling the coupling level. In this design, the spacing is approximately one-half of the trace width. Grounding the ground-line structure at both ends to form a loop helps achieve proper impedance matching while establishing the desired resonant frequencies. By appropriately optimizing the geometrical dimensions of the proposed structure, effective impedance matching is achieved across the three Wi-Fi operating bands, as illustrated in Figure 2a. In addition, the meandered loop ground-line structure functions as a planar sleeve balun, which helps reduce resonant-frequency variations

and improve the impedance bandwidth in the upper operating bands (5 GHz and 6 GHz). Figure 2b shows the prototype used as the experimental validation platform. In this implementation, the USB dongle PCB serves as the finite ground plane for the proposed antenna, representing a practical host-platform configuration for performance evaluation.



(**a**)



(**b**)

Figure 2. (**a**) Geometry and dimensions of the proposed antenna and the red circular indicator marks the explicit position of the feed point. (**b**) Photograph of the proposed antenna PCB inserted into the standard USB connector, showing the USB-connected experimental configuration used to establish the host-platform/reference-ground connection.

2.3. Ablation Study and Coupling Path Analysis

To systematically validate the underlying physics and individual contributions of the radiating and grounding elements in the proposed passive layout, a detailed ablation study and coupling path analysis were conducted.

2.3.1. Evolution of the Antenna Structure (Ablation Study)

First, to clarify the role of the peripheral grounding loop, three structural evolution stages were evaluated, as illustrated in Figure 3: the standalone T-type radiator (Figure 3a), the isolated meandered grounding loop (Figure 3b), and the fully integrated miniaturized antenna (Figure 3c). The simulated reflection coefficients ($|S_{11}|$) for these configurations are compared in Figure 4:

- As shown by the dashed curve, the standalone T-type radiator is inherently mismatched and fails to cover any of the targeted Wi-Fi bands.
- Similarly, the isolated grounding loop does not exhibit sufficient radiation or matching behavior on its own.
- However, when these two components are combined to form the fully integrated antenna, the structural synergy introduces a peripheral ground-coupled mechanism.

This framework successfully establishes the target tri-band operation, with the simulated $|S_{11}|$ dropping well below −10 dB across the 2.4 GHz, 5 GHz, and 6 GHz bands, proving that the interaction between the radiator and the meandered loop is vital for impedance matching.

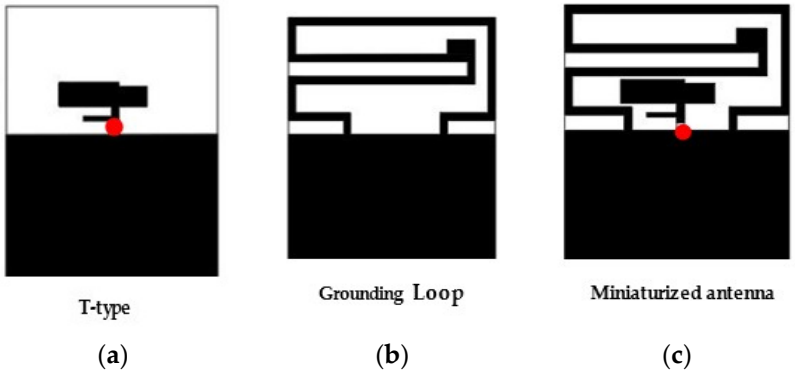


Figure 3. Antenna structural evolution for the ablation study: (**a**) Standalone asymmetric T-type monopole (ATM) radiator, and the red circular indicator marks the explicit position of the feed point. (**b**) isolated meandered grounding loop structure, and (**c**) proposed fully integrated miniaturized antenna.

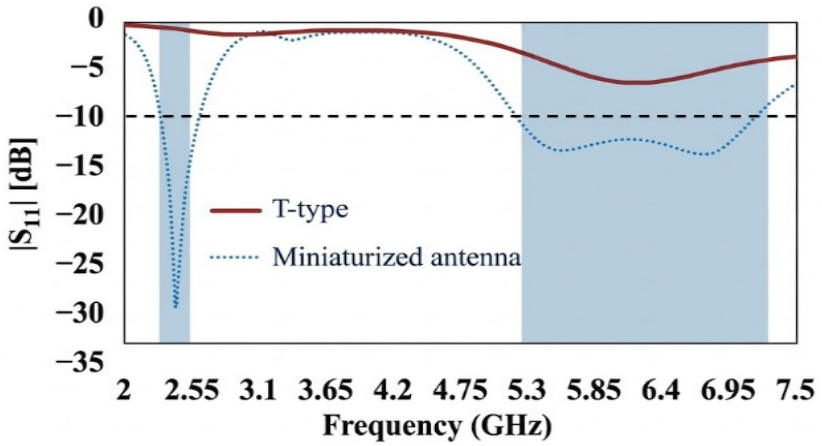


Figure 4. Simulated reflection coefficients $|S_{11}|$ comparing the standalone asymmetric T-type monopole (ATM) and the proposed integrated miniaturized antenna. The horizontal black dashed line represents the −10 dB impedance matching threshold, and the shaded regions indicate the targeted Wi-Fi operational bands.

2.3.2. Coupling Path Analysis Under Asymmetric Truncations

To further dissect the specific electromagnetic coupling paths between the ATM radiator and the peripheral meandered loop, two topological permutations involving asymmetric disconnections of the loop were analyzed:

- Case 1 (left-section disconnection, Figure 5b): The left section of the peripheral ground loop is truncated, leaving only the right path grounded. The corresponding simulated reflection coefficient in Figure 6 reveals that the multi-resonance characteristic vanishes, leaving only a single resonant mode. In this state, the primary monopole couples exclusively with the remaining ground segment, functioning as a basic loop antenna layout. This behavior is well-aligned with classical loop antenna theory but fails to meet the multi-band requirement.
- Case 2 (right-section disconnection, Figure 5c): The right section of the peripheral ground path is removed, leaving the left path connected. As plotted in Figure 6,

the simulated response for Case 2 exhibits three distinct resonant frequencies. In this state, the asymmetrical structural behavior closely resembles that of a conventional inverted-F antenna (IFA). However, due to the limited clearance and altered impedance path, the impedance bandwidth in the high-frequency bands is severely narrow and cannot satisfy the strict bandwidth criteria of Wi-Fi 6E/7.

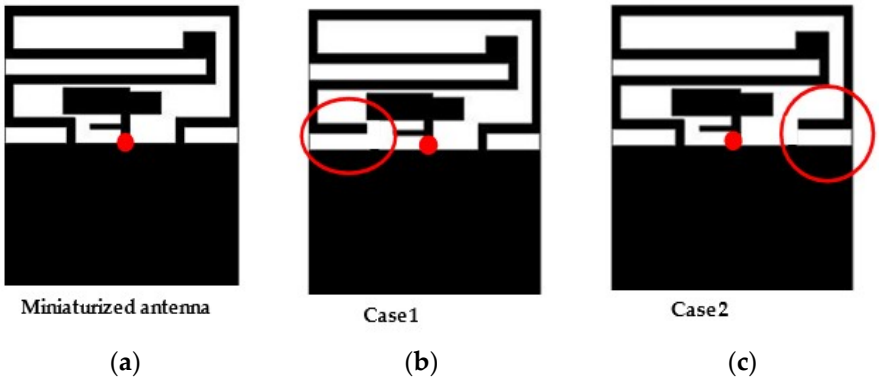


Figure 5. Evaluated topological permutations for coupling mechanism verification: (**a**) Proposed antenna with complete grounding loop path, and the red circular indicator marks the explicit position of the feed point; (**b**) Case 1 (truncated left path to evaluate single-loop mode); and (**c**) Case 2 (truncated right path to evaluate inverted-F antenna mode).

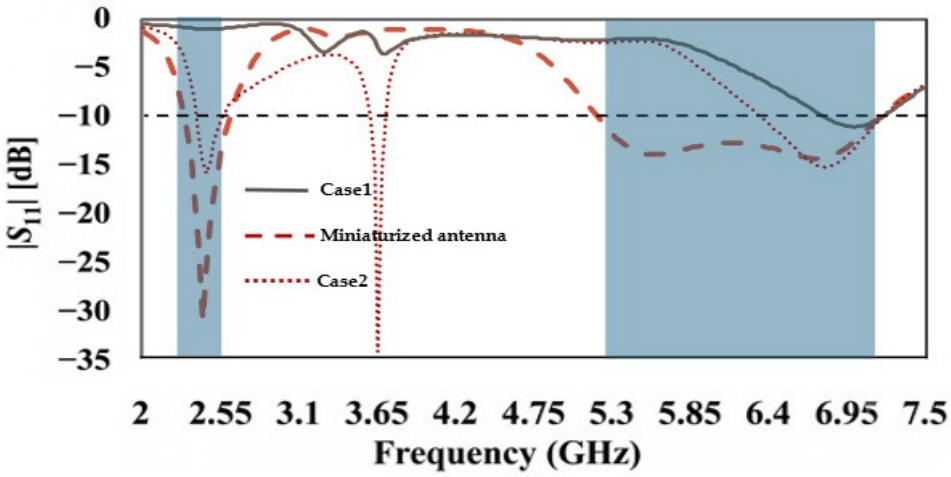


Figure 6. Simulated reflection coefficients $|S_{11}|$ for the coupling path analysis under different structural truncation cases (Case 1 and Case 2). The horizontal black dashed line represents the −10 dB impedance matching threshold, and the shaded regions denote the targeted Wi-Fi operational bands.

Although varying the dimensions or bending angles within these intermediate configurations (Case 1 and Case 2) can shift individual resonant frequencies, neither case can achieve the required broadband impedance matching simultaneously across all three bands. Ultimately, this comparative analysis underscores the paramount importance of the closed, dual-grounded meandered loop. The complete loop configuration provides the precise distributed reactive loading and balanced coupling paths necessary to generate the highly stable, tri-band resonances.

2.4. Simulation Antenna Current Distribution Analysis

Figure 7 illustrates the simulated surface-current distributions on the proposed antenna at distinct operating frequencies, providing profound physical insights into its multi-band excitation and coupling mechanisms. As shown in the Figure, the meandered-loop

ground wire wrapping around the ATM effectively splits the return surface currents into three distinct paths. At 2.45 GHz (Figure 7a), the surface current strongly couples to and distributes along the entire electrical length of the meandered-loop ground, demonstrating a characteristic one-quarter-wavelength resonance that establishes the lower band. At 5.5 GHz (Figure 7b), the near-field energy is predominantly concentrated on the outer ring profile of the geometry, validating the mid-band coupling mechanism. Conversely, at 6.5 GHz (Figure 7c), the surface-current density is highly concentrated along the primary branches of the ATM. This confirms that the asymmetric T-shaped monopole fundamentally operates as a standalone one-quarter-wavelength radiator to govern the highest frequency band. This joint optimization of shape coupling and alternative current routing proves that the lower bands are efficiently derived from the 6.5 GHz ATM foundation while maintaining excellent decoupling from external host perturbations.

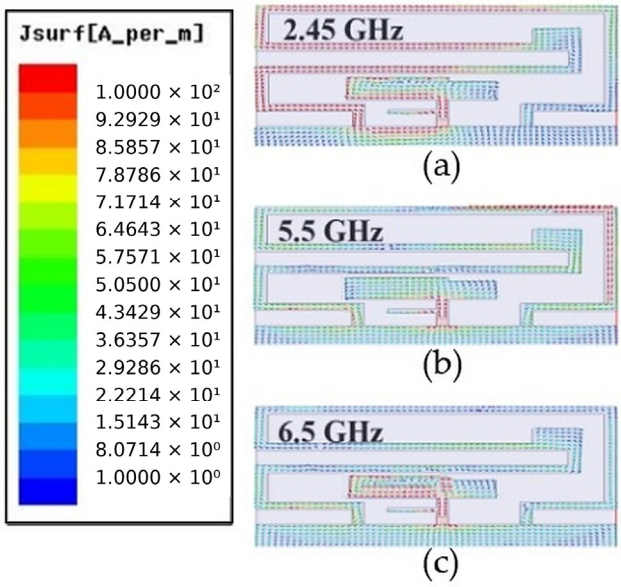


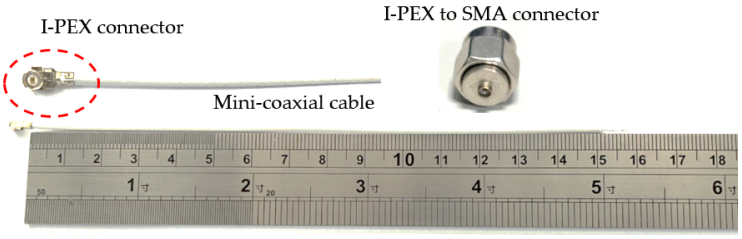
Figure 7. Simulated surface-current density plots on the conductive traces of the proposed antenna at the three core operating frequencies: (**a**) 2.45 GHz, (**b**) 5.5 GHz, and (**c**) 6.5 GHz (color scales indicate the current intensity in A/m).

3. Results and Discussion

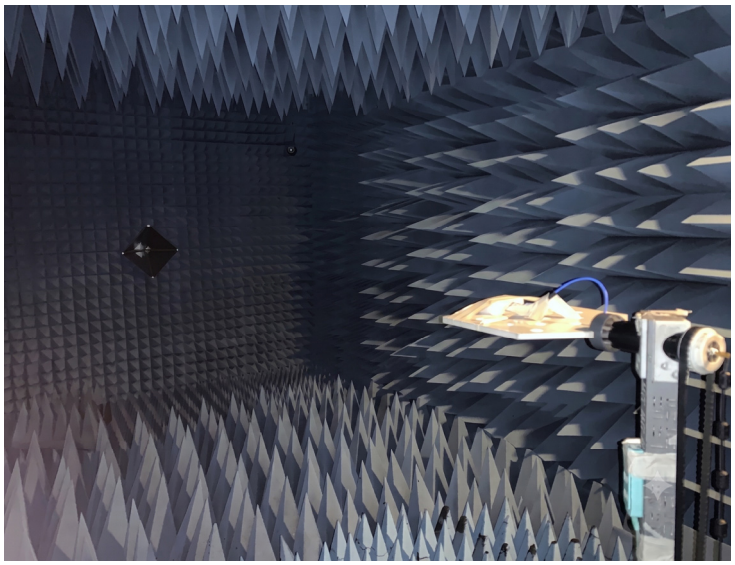
3.1. Measurement Environment Setup

The scattering parameters, specifically the reflection coefficient $|S_{11}|$, of the proposed USB dongle antenna were characterized using a Keysight E5071C Vector Network Analyzer (VNA) (Keysight Technologies, Santa Rosa, CA, USA) across the frequency range of 2.0 to 7.5 GHz. In order to ensure high measurement accuracy, a standard short-open-load-thru (SOLT) calibration procedure was executed prior to the characterization. For over-the-air (OTA) evaluation and RF characterization, a 150 mm-long, 1.3 mm-diameter 50 Ω mini-coaxial cable terminated with an I-PEX connector was employed as a temporary RF feeding interface. The I-PEX connector, the 150 mm-long mini-coaxial cable, and the I-PEX-to-SMA adapter are shown in Figure 8a. This coaxial assembly was used only during measurement and is not an integral part of the proposed antenna structure. The temporary feeding interface was introduced to provide a controlled 50 Ω RF connection to the measurement equipment. In the USB-connected configuration, the temporary coaxial cable is removed, and the antenna PCB is inserted into the standard USB connector, as shown in Figure 2b. The USB connector provides the physical and electrical interconnection between the antenna PCB and the host-platform/reference-ground structure, while the metal shield-

ing of the USB connector is electrically connected to the antenna ground plane. The far-field radiation parameters, including 2D/3D radiation patterns and antenna efficiencies, were characterized inside an ETS-Lindgren AMS-8500 OTA anechoic chamber (ETS-Lindgren, Cedar Park, TX, USA) under far-field boundary conditions. The physical infrastructure of the chamber encompasses dimensions of 7.32 m × 3.66 m × 3.66 m (L × W × H). As illustrated in Figure 8b, a Standard Double-Ridged Horn Antenna was employed as the reference measurement probe to transmit and receive the orthogonal polarization components. To ensure high measurement accuracy and minimize systematic uncertainties, a full-system calibration was performed prior to the over-the-air (OTA) measurements. The calibration accounted for free-space path attenuation, cable insertion losses, and I-PEX connector de-embedding. The RF insertion loss of the 150 mm-long, 1.3 mm-diameter, 50 Ω mini-coaxial cable was accurately characterized (approximately 0.29 dB at 2.45 GHz and 0.81 dB at 6.5 GHz) and subsequently de-embedded from the measured data to reduce the influence of the known feeding-path loss on the measured antenna response and OTA characterization. Furthermore, an integrated infrared positioning system was deployed inside the chamber to verify the precise alignment of the antenna's phase center during the automated spherical near-field/far-field scanning process.



(**a**)



(**b**)

Figure 8. (**a**) Detailed photograph of the temporary coaxial feeding interface, showing the I-PEX connector (red circle), the 150 mm-long mini-coaxial cable, and the I-PEX-to-SMA adapter used during antenna characterization (Note: The non-English character "寸" on the ruler denotes the traditional unit "cun", where 1 cun ≈ 3.33 cm; the primary reference scale is given by the metric centimeter/millimeter markings on the upper edge). (**b**) Experimental measurement configuration and antenna deployment inside the ETS-Lindgren AMS-8500 far-field OTA anechoic chamber.

3.2. Comparison of Simulated and Measured $|S_{11}|$

As illustrated in Figure 1a, the antenna structure is connected to a mini-coaxial testing cable with a 1.3 mm diameter and a length of 150 mm via a 50 Ω I-pex connector interface. Following the aforementioned design procedure, the triple-band antenna was fabricated for operation within the Wi-Fi spectrum allocations. Figure 9 depicts the simulated and measured $|S_{11}|$ curves of the proposed triple-band antenna. The measured impedance bandwidths satisfy the standard matching criterion of $|S_{11}| \leq -10$ dB over the span from 2.4 GHz to 2.5 GHz (with a percentage of 13.8% at 2.45 GHz), 5.15 to 5.85 GHz, and 5.925 to 7.125 GHz (with a percentage of 46% at 5.5 GHz and 37.38% at 6.5 GHz). The measured results demonstrate excellent agreement with the simulated predictions. In the experiments, the proposed antenna, with compact dimensions of 18 × 15 mm^2, was connected to a dummy motherboard serving as the reference ground-plane through a USB connector. The USB connector provided the electrical connection between the ground plane of the proposed antenna and the reference ground plane, as illustrated in Figure 10a. When the proposed USB dongle antenna is mounted on a 200 × 300 mm^2 reference ground plane, representing the motherboard of a laptop computer, a 15 cm-long, 1.3 mm-diameter, 50 Ω mini-coaxial cable terminated with an I-PEX connector is employed to measure the antenna reflection coefficient ($|S_{11}|$), as shown in Figure 10a.

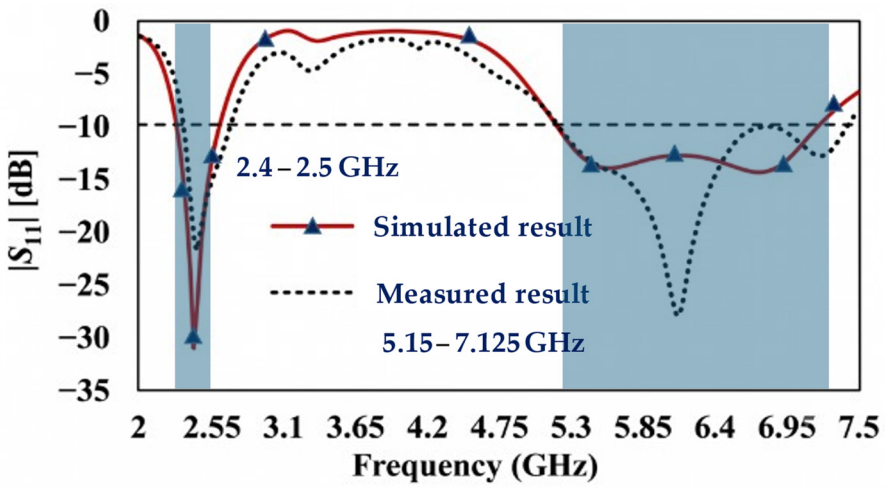
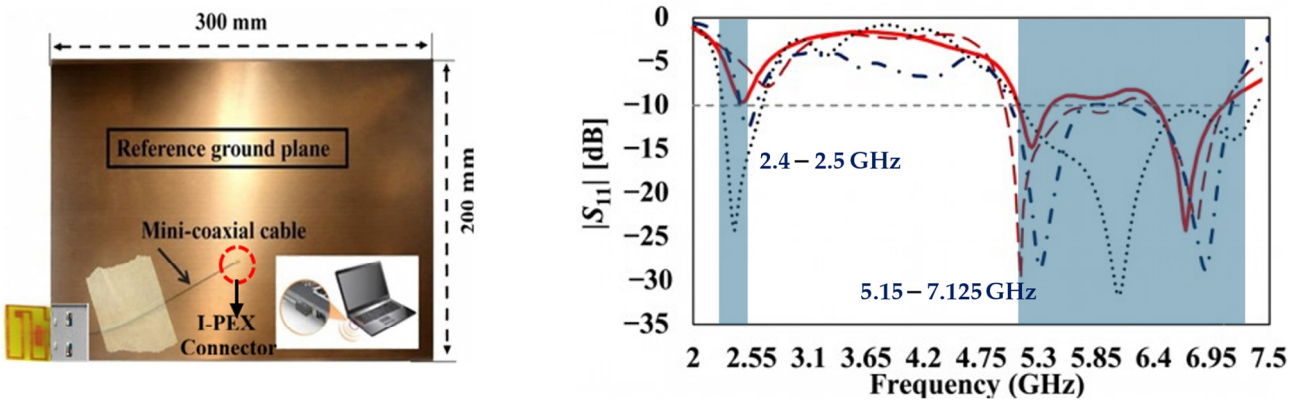


Figure 9. Comparison of simulated and measured reflection coefficients $|S_{11}|$ of the proposed standalone USB dongle antenna against frequency. The horizontal black dashed line represents the −10 dB impedance matching threshold, and the shaded regions indicate the targeted Wi-Fi operational bands.

As observed in the experimental multi-ground matrix curves in Figure 10b, the proposed antenna demonstrates consistent impedance performance across representative host-platform ground configurations. Although minor localized variations and a subtle impedance ripple around the 6 GHz band exist when migrating from the standalone scenario to the extremely large reference ground planes (e.g., 300 mm × 200 mm)—causing the commonly adopted $|S_{11}| \leq -10$ dB impedance-matching criterion at certain high-frequency band-edges—the principal tri-band resonant characteristics remain relatively stable. Crucially, from an electromagnetic perspective, the extreme stability of these measured peak frequencies under massive host–ground variations provide indirect experimental evidence supporting the proposed common-mode current suppression mechanism. In typical compact designs lacking decoupling elements, the host chassis acts as an uncontrolled radiating extension, leading to severe resonance shifts. Here, because the resonant peaks of the proposed design exhibit negligible frequency deviation even when connected to a reference ground plane approximately 400 times larger in area than the 10 × 15 mm^2

reference ground, the measured stability provides experimental evidence that the influence of the investigated host–ground variations on the ATM-based radiating structure is substantially reduced. This experimental behavior is consistent with the proposed planar sleeve balun mechanism discussed below. Guided by the calibration metrics established in Section 3.1, these subtle high-frequency deviations are identified as network-level measurement perturbations rather than an inherent failure of the decoupled antenna structure. These effects are considered potentially associated with several measurement-related factors, including:

- Manual soldering and assembly tolerances at the mini-coaxial feeding interface, which may introduce localized parasitic inductance and capacitance that become increasingly influential at higher frequencies.
- Residual common-mode coupling along the outer conductor of the extended mini-coaxial cable during cross-platform measurements, which may introduce additional perturbations in the measured phase and impedance response at the upper operating frequencies.



(**a**) (**b**)

Figure 10. Evaluation of antenna performance under representative finite-ground host-platform configurations: (**a**) Photograph of the fabricated USB-dongle antenna connected through the USB interface to a large reference ground plane representing a motherboard-scale host PCB, and (**b**) measured reflection coefficients ($|S_{11}|$) for different host–ground dimensions: dashed red line (---- 300 mm × 100 mm), dash-dot blue line (-·-·300 mm × 200 mm), solid red line (—150 mm × 200 mm). The horizontal black dashed line represents the −10 dB impedance matching threshold, and the shaded regions indicate the targeted Wi-Fi operational bands.

Because the present dataset does not allow these individual contributions to be independently separated, they are regarded as plausible measurement-related factors rather than quantitatively established individual causes. A complete feeding-assembly simulation or systematic de-embedded measurement would be required to isolate these effects. Nevertheless, the overall matching integrity and frequency alignment remain excellent, indicating reduced sensitivity of the antenna response to the investigated host–ground variations. When the proposed antenna is not connected to the mini-coaxial cable, it can be directly inserted into the USB connector, and the metal shielding of the USB connector is electrically shorted to the antenna ground plane. A photograph of the proposed antenna inserted into the USB connector is shown in Figure 2b.

As mentioned above, the proposed meandered-loop ground-line structure functions effectively as a modified planar sleeve balun. This mechanism is equivalent to connecting a one-quarter-wavelength short-circuited resonant loop to the reference ground plane, which presents an extremely high input impedance to common-mode currents, thereby

obstructing their path to the host platform. The simplified equivalent transmission-line model of the balun's suppression behavior is schematically illustrated in Figure 11.

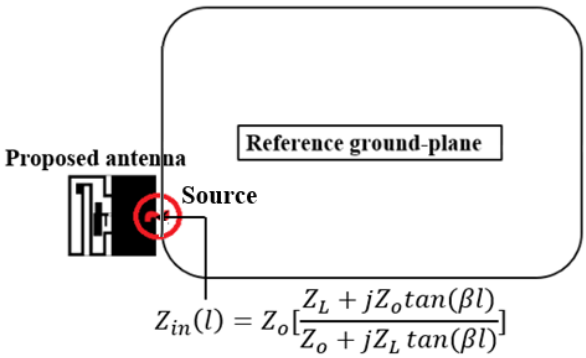


Figure 11. Equivalent transmission-line model of the proposed planar sleeve balun demonstrating the common-mode current suppression mechanism between the antenna and the massive reference ground plane.

Because both the start and end points of the meandered-loop ground line are tied directly to the ground plane, the terminating load impedance of this stub configuration can be mathematically represented as $Z_L = 0$. Under this boundary condition, the classic input impedance expression of a lossy transmission line in Equation (1) is simplified to Equation (2):

$$Z_{in}(l) = jZ_o tan(\beta l) \quad (2)$$

where Z_o represents the characteristic impedance of the microstrip line, β is the phase constant, and l represents the physical length of the path. When the total electrical length l of the loop corresponds to a quarter-wavelength (0.25λ) at the respective operating center frequency, the phase terms can be expressed as:

$$\beta l = \frac{2\pi}{\lambda} \times 0.25\lambda = \frac{\pi}{2}, tan(\beta l) \to \infty \quad (3)$$

Substituting Equation (3) into Equation (2) theoretically yields:

$$Z_{in}(l) = \infty \quad (4)$$

This high-impedance condition theoretically impedes the propagation path of common-mode currents, facilitating a stable operating bandwidth and suppressing frequency offsets in the upper bands.

It is worth noting that in a practical distributed microstrip layout, the meandered ground path cannot be treated as a completely lossless, ideal transmission line. In reality, non-ideal factors—such as localized capacitive and inductive parasitics introduced by the meandered right-angle bends, non-uniform surface-current distributions along the narrow traces, substrate dielectric losses, and near-field electromagnetic coupling with the adjacent ATM—will inevitably perturb the ideal impedance boundary. Consequently, the actual input impedance Z_{in} at the resonance node is finite rather than strictly infinite. Nonetheless, this simplified transmission-line model successfully captures the core physical wave mechanism of the design, generating a localized high-impedance barrier to isolate the radiator's reactive fields and significantly suppress common-mode current propagation back onto the host motherboard. Taken together, the ablation study, coupling-path analysis, simulated current distributions, equivalent transmission-line interpretation, and multi-ground measurements provide a consistent antenna-level explanation of the proposed mechanism. The ablation results demonstrate that the complete meandered loop is necessary to estab-

lish the desired tri-band response, while the current distributions identify the frequency-dependent current paths. The transmission-line model further explains the formation of a high-impedance condition along the meandered ground path, and the multi-ground measurements experimentally demonstrate that the resulting resonant characteristics remain relatively stable when the reference ground dimensions are substantially increased. These observations are mutually consistent with the proposed sleeve-loop mechanism. Nevertheless, because the common-mode current itself was not directly measured, the present evidence should be regarded as mechanism-supporting experimental evidence rather than a quantitative measurement of common-mode current suppression.

3.3. Comparison of Simulated and Measured 2D/3D Radiation Patterns

Figure 12 presents the measured 3D and 2D radiation patterns at 2.45 GHz, 5.5 GHz, and 6.5 GHz, respectively. The measured radiation patterns exhibit multilobe characteristics at 5.5 GHz and 6.5 GHz, which can be attributed to the excitation of undesired higher-order modes of the meandered loop ground line at higher frequencies, resulting in distorted radiation patterns. As plotted in Figure 12a, the proposed antenna exhibits a quasi-omnidirectional radiation characteristic in the XY-plane at 2.45 GHz, which is highly desirable for mobile Wi-Fi terminal applications to ensure stable coverage. At 5.5 GHz and 6.5 GHz (Figure 12b,c), although higher-order resonant modes introduce multilobe radiation patterns at 5.5 GHz and 6.5 GHz, the measured three-dimensional radiation distributions maintain spatially distributed radiation characteristics, although the multilobe behavior becomes more pronounced at the higher frequencies. A quantitative coverage-efficiency or minimum-realized-gain metric was not included in the present study and is therefore not used to make a quantitative claim regarding three-dimensional coverage performance. This limitation is further discussed in Section 3.5. Table 2 shows the peak gains and 3D pattern efficiency measurements of the antenna within the operating bands. The antenna in the proposal demonstrates that the measured gain and radiation efficiency characterize the radiation performance at the investigated frequencies.

Table 2. Measured peak gains and radiation efficiencies of the proposed antenna at key operating frequencies.

Frequency (GHz)	Peak Gain (dBi)	Radiation Efficiency (%)
2.4	2.79	52
2.45	3.26	53
2.5	3.73	58
5.15	4.76	58
5.5	4.42	88
5.85	6.07	77
5.925	5.62	69
6.5	4.59	62
7.125	2.89	48

3.4. Performance Comparison with State-of-the-Art Works

In order to further contextualize and highlight the performance bounds of the proposed design, a comprehensive comparison with recently published state-of-the-art Wi-Fi 6E/7 multi-band antennas is summarized in Table 3. To ensure a rigorous and transparent comparison, and to explicitly address the concern regarding disparate test configurations, key parameters—such as antenna radiator footprint, host–ground dimensions, substrate material, structural complexity, matching criteria, and platform decoupling mechanisms—are systematically categorized. As tabulated, several prior counterparts rely on relaxed impedance matching criteria ($|S_{11}| \leq -6$ dB or -7.4 dB) or discrete lumped components

to establish multi-band operation within small clearances [7,8,16,17]. Furthermore, highly integrated topologies tailored for laptop frames or mobile platforms often require complex 3D multi-facet Laser Direct Structuring (LDS) structures, multi-layered absorber shielding, or specific metal-frame configurations to suppress platform interference [6,7,18]. Other representative planar tri-band configurations, such as the metamaterial-loaded monopole [21] and the dual-monopole structure [22], achieve impedance matching by etching defected ground structures (DGS) on the chassis. However, these designs require relatively large radiator clearances (20.4×26 mm^2 in [21] and 19.3×17.1 mm^2 in [22]) and remain limited to legacy sub-6-GHz WLAN/WiMAX allocations without covering the upper 6 GHz Wi-Fi 6E/7 spectrum (up to 7.125 GHz). In stark contrast, despite occupying an ultra-compact radiator clearance of only (8×15 mm^2) alongside a minimal reference ground footprint (10×15 mm^2), the proposed antenna rigorously satisfies the standard industrial threshold of $|S_{11}| \leq -10$ dB across its core bands under standalone conditions, while preserving stable frequency alignment under massive multi-ground extensions. Backed by the co-designed planar meandered sleeve-loop balun, the proposed single-sided distributed layout achieves an exceptional maximum radiation efficiency of 88% and a peak gain of 6.07 dBi across all three Wi-Fi bands without requiring discrete lumped components, external tuning networks, 3D LDS carriers, or etched ground defects, validating its structural simplicity, compactness, and robust platform insensitivity.

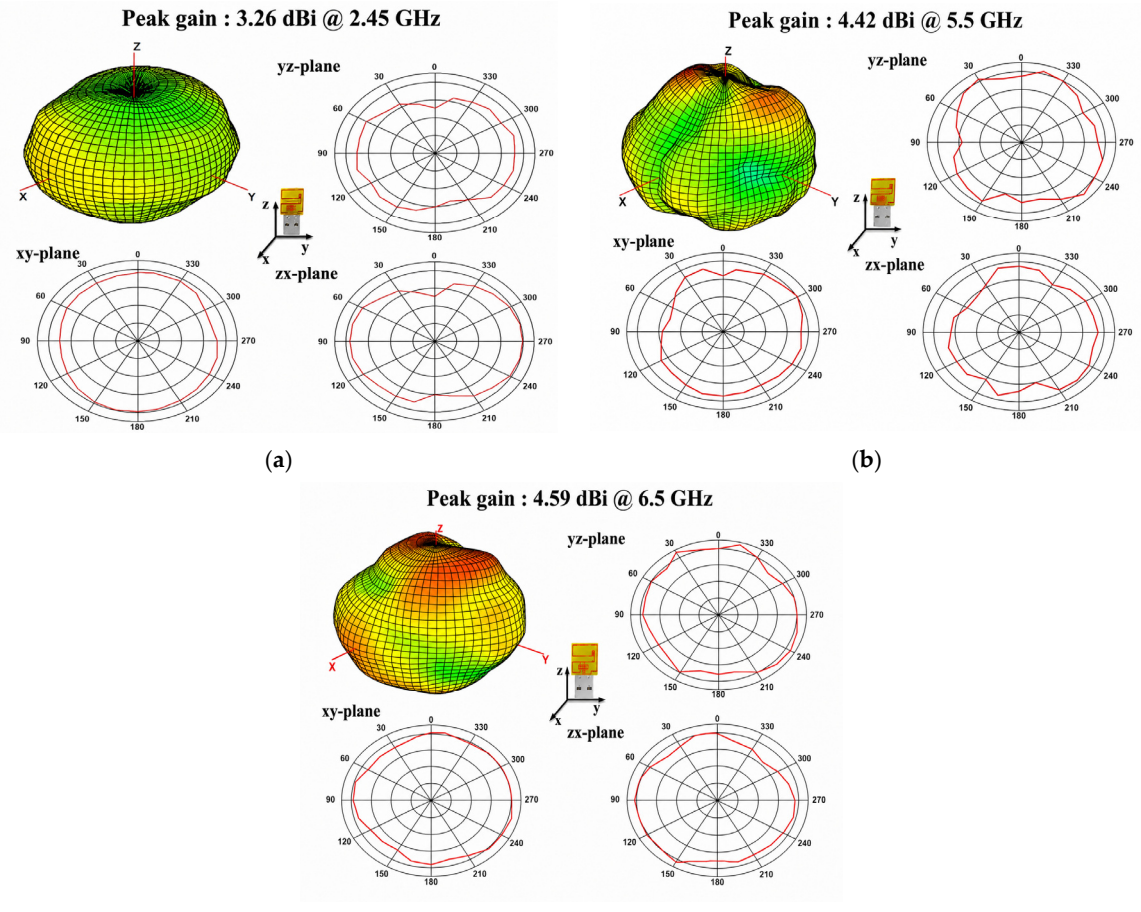


Figure 12. Measured 3D and 2D polar radiation patterns of the proposed antenna at the three core operating frequencies: (**a**) 2.45 GHz, (**b**) 5.5 GHz, and (**c**) 6.5 GHz, where the 3D surface color mapping indicates normalized radiation intensity (red corresponds to maximum radiation/peak gain, while green indicates lower radiation levels), and the solid red lines in the 2D polar plots represent the total radiation patterns across the principal cut planes (x–y, x–z, and y–z planes).

Table 3. Performance comparison between the proposed antenna and recently reported state-of-the-art Wi-Fi 6E/7 antennas.

Reference	Antenna Area (mm^2)	Ground-Plane Size (mm^2)	Operating Band (GHz)	Peak Gain (dBi)/Max. Radiated Efficiency (%)	Matching Criterion	Tuning/Matching Components	Platform Decoupling Mechanism	Structural Complexity	Substrate
[6] 2026	34.1 × 4.6	50 × 10	2.40–2.48/ 5.15–7.125	6.3/80%	$\lvert S_{11}\rvert \leq -10$ dB	Multi-Segment Strips	Unified RF-Module Absorber Shielding	Multi-Layer with Absorber (Complex)	FR4 (ε_r = 4.4/ 0.8 mm)
[7] 2023	8 × 30	180 × 310	2.40–2.48/ 5.15–7.125	5.7/60%	$\lvert S_{11}\rvert \leq -6$ dB	Stubs & Parasitic Strip	Three-Sided Vertical Ground Walls	3D Multi-Facet (Complex)	LDS (ε_r = 2.9/ 8 mm)
[8] 2025	6 × 8	40 × 80	2.20–2.65/ 4.81–7.45	3.73/60.7%	$\lvert S_{11}\rvert \leq -6$ dB	Chip Capacitors & Inductors	Ground Radiation Technique	Single-Layer with SMD (Medium)	FR4 (ε_r = 4.4/ 1.0 mm)
[16] 2021	4 × 16	1 × 5	2.4–2.5/ 5.15–7.125	5.7/70%	$\lvert S_{11}\rvert \leq -7.4$ dB	3 Chip Inductors	Symmetrical Inductive Shorting	Planar with SMD (Medium)	FR4 (ε_r = 4.4/ 0.8 mm)
[17] 2021	6 × 30	182 × 315	2.40–2.48/ 5.15–7.125	6.0/86%	$\lvert S_{11}\rvert \leq -7.4$ dB	Meandered Tuning Portion	Offset-Fed Self-Isolation	Single-Layer Planar (Simple)	FR4 (ε_r = 4.4/0.8 mm)
[18] 2025	25.7 × 1.2	210 × 310	2.33–2.49/ 4.92–7.32	~2.5/70%	$\lvert S_{11}\rvert \leq -10$ dB	Step Slot & Loop Slit	Small Clearance Slot Resonance	Double-Sided Metal Frame (Medium)	FR4 (ε_r = 4.3/ 0.8 mm)
[21] 2021	20.4 × 26	8.6 × 26	2.38–2.48/ 3.37–3.79/ 4.36–6.06	3.66/NA	$\lvert S_{11}\rvert \leq -10$ dB	Metamaterial Unit Cell	Annular Defected Ground	Double-Sided DGS (Medium)	FR4 (ε_r = 4.4/ 1.6 mm)
[22] 2020	19.3 × 17.1	9.3 × 39	2.35–2.52/ 3.2–4.16/ 5.13–5.87	4.06/95%	$\lvert S_{11}\rvert \leq -10$ dB	Concentric Ring-Patch	Arc-Shaped Defected Ground	Double-Sided DGS (Medium)	FR4 (ε_r = 4.4/ 1.6 mm)
Proposed antenna	**8 × 15**	**10 × 15**	**2.4–2.5/ 5.15–5.85/ 5.925–7.125**	**6.07/88%**	**$\lvert S_{11}\rvert \leq -10$ dB (Core bands) ***	**None (Distributed Layout)**	**Planar Meandered Sleeve-Loop**	**Single-Sided Planar (Simple)**	**FR4 (ε_r = 4.3/ 1.0 mm)**

* Note: Minor localized deviations occur at the extreme high-frequency band-edges under massive reference ground hosting potentially associated with practical feeding and interconnection effects.

3.5. Limitations and Scope of This Work

The present study focuses on the development of a compact tri-band antenna architecture and the experimental investigation of its robustness against representative variations in host-platform ground configurations. The primary objective is to address resonance detuning and impedance instability associated with highly miniaturized antennas operating with substantially different finite ground dimensions. Within this scope, the measured results demonstrate relatively stable resonant characteristics and impedance response across the representative host–ground configurations investigated in this work. Nevertheless, several aspects remain beyond the scope of the present study and should be acknowledged:

- First, the proposed common-mode current suppression mechanism is supported by simulated surface-current distributions, structural ablation, coupling-path analysis, the equivalent transmission-line model, and measured resonance stability under different host–ground configurations. However, direct quantitative extraction of common-mode current magnitude, current-suppression ratio, or characteristic-mode parameters was not performed. Similarly, the host–ground investigation employed controlled finite-ground configurations to isolate the effect of ground-plane dimensions rather than complete commercial platforms. Therefore, the present results do not comprehensively account for additional effects associated with different platform shapes, materials, connector locations, nearby components, chassis structures, or device housings. Quantitative current-based analysis and realistic platform-level investigations are left for future work.
- Second, the antenna was characterized using a temporary mini-coaxial cable and I-PEX connector. Although feeding-path loss and connector-related effects were considered during the measurement process, a complete parametric investigation of feeding-assembly parasitics was not conducted. The higher-frequency deviations observed near certain band edges may be affected by connector, soldering, cable-routing, and residual common-mode effects. These deviations are therefore treated as possible measurement-related effects rather than being attributed exclusively to a single physical cause. A fully integrated feeding structure and complete feeding-assembly simulation would provide further validation.
- Third, the measured radiation patterns demonstrate the spatial radiation characteristics of the antenna at representative frequencies, with relatively stable behavior at 2.45 GHz and more pronounced multilobe characteristics at 5.5 and 6.5 GHz due to higher-order resonant modes. However, quantitative metrics such as coverage efficiency, minimum realized gain, cumulative spatial gain distribution, and complete co-/cross-polarization characteristics were not included in the present study. Accordingly, no quantitative claim is made regarding worst-case orientation performance or polarization purity over the entire operating bandwidth. In addition, a systematic simulated-versus-measured 3D radiation-pattern comparison was not available for all representative frequencies.
- Finally, the present investigation considers a single antenna element. Although Wi-Fi 7 products commonly employ multi-antenna MIMO architectures, the primary objective of this work is to establish and experimentally validate the proposed single-element radiator–sleeve-loop architecture and its robustness against host–ground variations. Therefore, mutual coupling, envelope correlation coefficient (ECC), total active reflection coefficient (TARC), and multi-element isolation were not investigated. The proposed antenna is considered a compact building block that may be extended to multi-port configurations, while complete MIMO validation remains beyond the scope of the present study.

These limitations do not affect the primary objective of this work, which is to demonstrate a compact tri-band antenna architecture with improved robustness against representative host–ground variations. The experimental results support this objective through the observed stability of the resonant characteristics, impedance response, and measured radiation performance under substantially different finite-ground configurations. Quantitative common-mode-current characterization, realistic platform integration, complete feeding-assembly modeling, spatial gain statistics, polarization-resolved measurements, simulated/measured 3D pattern comparison, and multi-element MIMO evaluation are considered important directions for future work.

4. Conclusions

This paper presents the design and experimental validation of a compact tri-band antenna engineered for Wi-Fi 6E and Wi-Fi 7 applications. Occupying a highly miniature footprint of only $8 \times 15\ mm^2$, the antenna incorporates an asymmetric T-shaped monopole (ATM) combined with a meandered loop ground-line structure to achieve multi-band operation covering the 2.4, 5, and 6 GHz WLAN bands. Acting as a modified planar sleeve balun, the meandered loop is designed to reduce PCB-induced common-mode current propagation and consequently mitigate the influence of the host–ground plane on the antenna characteristics. Experimental results demonstrate that the proposed antenna maintains relatively stable resonant frequencies and impedance characteristics across the representative host–ground configurations investigated in this work, thereby reducing the frequency detuning and impedance degradation commonly encountered in compact antennas when the host–ground configuration is changed. The observed stability supports the effectiveness of the integrated radiator–sleeve-loop configuration in reducing the sensitivity of the antenna to host–ground variations within the investigated test conditions. Broadband impedance matching is achieved through optimized electromagnetic coupling between the ATM radiator and the meandered loop, resulting in a simple planar antenna architecture without additional lumped matching components. Regarding radiation performance, the antenna exhibits a quasi-omnidirectional radiation characteristic at 2.45 GHz, while multilobe behavior becomes more pronounced at 5.5 and 6.5 GHz due to the excitation of higher-order modes associated with the meandered ground-loop structure. The measured 3D radiation distributions maintain spatially distributed radiation characteristics, although multilobe behavior becomes more pronounced at the higher frequencies. The measured peak gain and radiation efficiency further demonstrate the radiation performance of the proposed antenna at the selected representative frequencies. However, a quantitative coverage-efficiency or minimum-realized-gain metric was not included in the present study; therefore, the practical impact of the observed high-frequency radiation-pattern variations on orientation-dependent coverage cannot be quantitatively established from the present measurements. This aspect is identified as a limitation of the current work and will be investigated in future studies. The scope of the present work is intentionally focused on the antenna-level mechanism and its robustness against representative finite host–ground variations, rather than on complete commercial-device validation. In particular, the investigated host platforms are controlled finite-ground configurations and do not encompass all possible combinations of platform geometry, dielectric or housing materials, connector locations, nearby electronic components, or complete product enclosures. Furthermore, the proposed antenna is demonstrated as a single antenna element, and therefore multi-antenna Wi-Fi 7 characteristics, including mutual coupling, isolation, envelope correlation coefficient, and total active reflection coefficient, have not been experimentally investigated in this study. Compared with existing miniaturized antennas, the proposed architecture offers a compact radiator clearance, tri-band operation covering the principal

Wi-Fi 6E/7 frequency allocations, and an integrated distributed matching and host–ground-current suppression mechanism without additional lumped matching components. The main advantage of the proposed design is therefore its demonstrated robustness against the representative host–ground variations investigated in this work, rather than a claim of universal platform independence or complete Wi-Fi 7 system-level applicability.

Furthermore, the planar structure can potentially be implemented using single-sided flexible printed circuit (FPC) or Laser Direct Structuring (LDS) technologies, facilitating integration into compact wireless devices. Although this work focuses on a USB-dongle-based experimental platform, the proposed planar sleeve-loop concept is not intrinsically limited to the USB interface and may serve as a compact antenna building block for other space-constrained wireless platforms. Future work will investigate the extension of the proposed antenna concept to more realistic product-level environments and multi-antenna (MIMO) Wi-Fi 7 systems, including different platform geometries and materials, connector and component effects, complete housing configurations, mutual coupling, isolation, envelope correlation coefficients, total active reflection coefficients, and quantitative 3D coverage metrics such as minimum realized gain and coverage efficiency.

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